



ROUND 2

Instructions and Information

1. This is a **7-question, 2-day test**, in which all of them are **proof** questions. Each question has a given weight (number of points) shown in the test beside the problem. The maximum number of points you can get is 50.
2. For all questions, please **scan your papers** and submit it as **separate PDFs** to your assigned Discord channel and ping mathemolympiad.
3. **No calculators**, books, online resources, etc. are allowed, and no other tab aside from the test paper and the Discord contest channel should be opened. Only blank scratch paper, ruler, compass, and writing materials are allowed. If anyone is caught with illegal resources, they will be disqualified.
4. No person who participated or helped in the contest is allowed to discuss any aspect of it **48 hours from the contest date**. If moderators or trial moderators catch anyone discussing during this time frame, they will automatically be disqualified.
5. Ties among two teams will be broken according to the team who scores higher in the **last problem that their scores differ**, then further ties will be broken according to their score in Round 1.

Problems

1. [6] Let p be a prime. I chose a secret permutation $a_0, a_1, \dots, a_{p-1}$ of $0, 1, \dots, p-1$ unknown to you. Now, you can ask the following types of questions to me:
 - Type 1: Tell me two integers i, j where $0 \leq i, j \leq p-1$. I will then tell you the k such that $a_k \equiv a_i + a_j \pmod{p}$.
 - Type 2: Tell me two integers i, j where $0 \leq i, j \leq p-1$. I will then tell you the k such that $a_k \equiv a_i a_j \pmod{p}$.

You may ask at most $p-1$ questions. Show that you have a strategy to determine the correct permutation after asking the questions.

2. [6] Let n be an integer and x be a real number. Consider $a_1, a_2, \dots, a_n$ being a permutation of $1, 2, \dots, n$. Furthermore, recursively define $f_1(k) = a_k$ and $f_{i+1}(k) = a_{f_i(k)}$ for all positive integers k, i where $k \in \{1, 2, \dots, n\}$. Define the score of a permutation be the maximal possible size of a set S being a subset of $\{1, 2, \dots, n\}$ such that there does not exist two distinct elements p, q of S and an integer m such that $f_m(p) = q$. Find the sum over all permutations of length n for x to the power of its score.
3. [7] Let O be the point of intersection of the diagonals of the cyclic quadrilateral $ABCD$. The circumcircles of $\triangle ABO$ and $\triangle CDO$ intersect at point K , different from point O . If the point L is taken outside the convex quadrilateral $ABCD$ such that $\triangle BLC \sim \triangle AKD$, prove that there exists an incircle for the quadrilateral $BLCK$.
4. [7] There are four icosahedral fair dice A, B, C, D . Dice A and B each has its faces numbered from 1 to 20 respectively. Dice C and D has a positive integer on each of its faces, such that for any integer n , the probability of getting a sum of n by rolling dice A and B is equal to the probability of getting a sum of n by rolling dice C and D . Find the minimum possible value of the largest number on dice C .
5. [7] For all positive real numbers x, y, z satisfying $x + y + z = 2$, prove that

$$\frac{(x-1)^2}{y} + \frac{(y-1)^2}{z} + \frac{(z-1)^2}{x} \geq \frac{1}{4} \left(\frac{x^2+y^2}{x+y} + \frac{y^2+z^2}{y+z} + \frac{z^2+x^2}{z+x} \right)$$

6. [8] Let $n \geq 3$ be a positive integer. A country consists of n islands. To attract tourists, the government of the country decided to set up n two-way tourism ship routes, in which each ship route connects two islands in the country, and one can travel between any two islands in the country only using the tourism ship routes. Now, you, being a government official, is tasked to give a *rating* of 1 or 2 to each of the n tourism ship routes. The *attractiveness* of an island is the sum of all the ratings of the ship routes connecting to that island. A restriction is that the attractiveness of two islands directly connected by a tourism ship route has to be different. Find all configurations of the tourism ship routes such that you can assign a rating to all tourism ship routes satisfying the restriction.

7. [9] There are three distinct lines l_1, l_2, l_3 concurring at X . Lines l'_i for $1 \leq i \leq 3$ and $A_{i,j}$ for $1 \leq i, j \leq 3$ are constructed such that $l_i \perp l'_i$ for $1 \leq i \leq 3$, l_i and l'_j intersect at $A_{i,j}$ for $1 \leq i, j \leq 3$, and $A_{1,1}, A_{2,2}, A_{3,3}$ are collinear. Then, for each of $1 \leq i \leq 3$, we do the following (indices modulo 3):

- Let D_i be the intersection of l'_i and line $A_{i-1,i+1}, A_{i+1,i-1}$.
- Let S_i be the intersection of the circumcircle of $\triangle XA_{i-1,i+1}A_{i+1,i-1}$ and line XD_i other than X .
- Let M_i be the midpoint of $A_{i-1,i+1}$ and $A_{i+1,i-1}$.
- Let C_i be the intersection of the perpendicular bisector of XS_i and the line through M_i parallel to l_i .

Prove that C_1, C_2, C_3 are collinear.

Marking Schemes for Selected Problems

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PI marking

Scores are cumulative

Solve special case of $p=2$

— 1 mark

Remaining cases

Use first type of question on 0 0 for first question then

split into cases according to if response is 0 or not

— 1 mark

Find the ratios of $a_0, a_1, \dots, a_{p-1}$ in $p-3$ questions for

- The response of first question is not 0

- The response of first question is 0

} Both: 3 marks
One: 1 mark

Use the second type of question to find the permutation
from the ratios — 1 mark

P2 marking

Marks are cumulative

Answer: $x(x+1)\dots(x+n-1)$

—— 1 mark

Identifying score = number of cycles

—— 1 mark

Briefly explain above

—— 1 mark

Induction or any way to get final answer

—— 3 marks

P4 marking

Answer: 8

— 1 mark

Apply generating function

- Dice A & B $\rightarrow x+x^2+\dots+x^{20}$

— 1 mark

- Correctly state the new translated problem

— 1 mark

- Factorise $x+x^2+\dots+x^{20}$ into polynomials irreducible over $\mathbb{Q}$ and state why they are irreducible (cyclotomic polynomials)

— 1 mark

- Use the sum of coefficients (or equivalently, the value at $x=1$) to impose a constraint on the factors of the GF of dice C and D

— 1 mark

- Show that 7 does not work due to negative coefficients

— 1 mark

- Show that 8 works

— 1 mark

P6 marking

Answer: Only cycles of length not being a multiple of 4 have no valid assignment. — 1 mark

State the graph consists of exactly 1 cycle — 1 mark

Solve the case where whole graph is cycle — 1 mark

Show if there is a valid assignment for the cycle vertices, then remaining vertices can be validly assigned — 1 mark

Solving the remaining case — 4 marks

In particular, the following will get partials from the 4 marks

- Show that a valid construction exists if a cycle vertex has degree at least 4 — 1 mark